

Bryan Smith

Data Scientist & Data Engineer

Data Analytics • Business Intelligence • Software Engineering

Azure • Microsoft Fabric • Spark • Python • SQL

Pretoria, South Africa

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Summary

Data Scientist and Data Engineer who builds and owns production data platforms. At Peregrine Capital I built the platform that runs the business's analytics: 100+ pipelines drawing from dozens of source systems into hundreds of modelled tables, feeding an enterprise reporting estate used firm-wide. Hands-on across Azure, Microsoft Fabric, Spark, Airflow, Python, and SQL, with depth in data ingestion, warehousing, schema design, database administration, and performance tuning, plus applied machine learning and generative AI on top. I adapt quickly. I pivoted from actuarial and financial mathematics into data, then learned each new stack on the job, from Ab Initio and Teradata at First National Bank through Azure, Microsoft Fabric, and AI tooling at Peregrine. I have since grown into technical leadership and software engineering: mentoring and onboarding team members, setting technical direction, and building internal web applications and tooling. My delivery standards are pragmatic: secure access patterns, automated deployments, and proactive alerting. This helps teams move fast without trading away reliability.

Experience

Data Scientist

Peregrine Capital

Sandton

Nov 2021 – Present

Built and own Peregrine's end-to-end data platform: 100+ production pipelines drawing from dozens of source systems into hundreds of modelled tables, feeding an enterprise reporting estate used across the business.

- Built the platform from the ground up, initially using Airflow, Docker, and Postgres before migrating to Microsoft Fabric Pipelines, PySpark, and T-SQL. It now runs 100+ pipelines across hundreds of modelled tables. Own platform-level architecture and administration.
- Unified dozens of source systems into a single analytics ecosystem, spanning file transfers, APIs, and web-based integrations, including sources with no API. Also built and maintained functionality within a CRM platform used as a system of record for the analytics team.
- Replaced schedule-based file ingestion with event-driven, stream-based processing, triggering downstream work the instant a file lands and removing the guesswork of scheduling around estimated arrival times.
- Built unsupervised models that flag data points falling outside a dataset's expected norm, and engineered features for downstream analysis including temporal decomposition and derived labels.
- Delivered a wide-reaching enterprise reporting estate across Power BI and paginated reports, now used firm-wide, owning both the visualisation layer and its governance.
- Designed and maintained logical and physical data models using Kimball methodology with Snowflake schemas, and owned database design and administration across reporting and analytics environments, including schemas, access controls, and performance tuning.
- Embedded LLMs into operational processes via Azure AI Foundry: OCR paired with LLM classification to turn document images into structured data, and speech-to-text plus LLM extraction that turns meeting discussions into tracked, documented work items. Also built an LLM-powered spam detection agent for inbound messages, a query router agent that directs user requests to the appropriate tool, workflow, or specialist agent, and AI-assisted developer tooling to support the team's day-to-day engineering work.
- Implemented pipeline- and report-level alerting that flags data quality issues before invalid outputs reach reports, backed by an AI agent that assists with error diagnosis and remediation guidance.
- Built secure internal web applications on Azure Container Apps and Azure Static Web Apps, plus automation using Azure Logic Apps, Function Apps, Storage Accounts, and Key Vault. Delivered dashboards for pipeline control, data access, and notifications with enterprise authentication and secrets management.
- Work directly with stakeholders from requirements analysis and solution design through architectural planning, environment setup, implementation, deployment, and support. Also provide mentorship, code review, onboarding, and technical direction, managing delivery through Azure DevOps with structured environments and formal change control.

Built high-performance ETL pipelines and large-scale data integrations in an enterprise banking environment.

- Built dynamic, reusable ETL pipelines using Ab Initio and Teradata SQL, ingesting from Kafka queues, Postgres, and MySQL to power the App, USSD, and Online channels.
- Redesigned Teradata table structures to balance cost against speed through strategic indexing, structural refinements, and multi-value compression (MVC), cutting storage by up to 50%.
- Tuned pipelines to process gigabytes within short batch windows through repeated rounds of indexing, partitioning, and resource management.
- Developed logical and physical data architectures spanning dozens of interconnected tables.
- Managed SQL environment permissions and access controls across production and development layers, minimising risk.
- Progressed rapidly into code review and mentorship responsibilities, trusted by senior engineers to deliver production-grade code.

Education

BSc Computer Science (Part-Time) 2022 – 2025
University of Witwatersrand, Johannesburg
Relevant coursework: Machine Learning, Parallel Computing, Advanced Analysis of Algorithms, Software Design.

BSc Actuarial and Financial Mathematics 2016 – 2019
University of Pretoria, Pretoria
Relevant coursework: Multivariate Analysis, Stochastic Processes, Time Series Analysis, Financial Engineering.

Actuarial Exams (ASSA)
Actuarial Society of South Africa

Skills

Data Processing: Spark, Batch Processing, Stream Processing, Ab Initio
Cloud: Azure Container Apps, Azure Static Web Apps, Azure Functions, Azure Key Vault, Azure Logic Apps
Data Ingestion & Source Systems: API Ingestion, SFTP Ingestion, Salesforce, SharePoint, Web Scraping
Data Platforms & Warehousing: Microsoft Fabric, Microsoft SQL Server, Postgres, MySQL, Teradata
Data Modeling & Database Management: Dimensional Modeling (Kimball), Database Design, Database Administration
AI & Machine Learning: Azure AI Foundry, LLM Integration, Agentic AI Development, Unsupervised Learning, Predictive Modeling
Business Intelligence: Power BI Reports, Power BI Paginated Reports, DAX
Analytics & Modeling: Statistical Analysis, Feature Engineering, Data Visualization
Infrastructure & DevOps: Azure DevOps, Monitoring & Alerting, Docker, Airflow, CI/CD
Programming & Scripting: Python, SQL (T-SQL), Bash, R
Software Engineering: JavaScript, React

Languages

Afrikaans (Fluent), English (Fluent)